

何铭诚 (MINGCHENG HE)

滑铁卢大学
电子与计算机工程

博士后研究员
mingcheng.he@uwaterloo.ca

研究方向

空天地一体化网络

卫星地面网络辅助低空技术研究
大规模卫星网络建模、性能评估、智能调度与协议设计
异构网络下中资源切片设计，通信、计算、缓存等多维度资源管理
数据驱动的服务需求预测与流量感知

学术研究经历

博士后研究员

2025 年 2 月 – 至今

宽带通信研究 (BBCR) 实验室, 滑铁卢大学, 加拿大
导师: 沈学民教授 Xuemin (Sherman) Shen、庄卫华教授 Weihua Zhuang

- 空天地网络、人工智能、数字孪生研究
 - 面向低空应用的卫星地面算力网络
 - 智能体和数字孪生赋能星地网络资源调度
 - 基于人工智能技术的资源切片技术

研究助理

2020 年 9 月 – 2024 年 12 月

宽带通信研究 (BBCR) 实验室, 滑铁卢大学, 加拿大

- 星地融合车联网络研究
 - 星地融合车联网络网络性能分析
 - 制定星地融合车联网络多维资源调度方案
 - 研究星地一体化车载网络资源切片人工智能技术

教育经历

博士, 电子与计算机工程

2020 年 9 月 – 2024 年 12 月

滑铁卢大学, 加拿大, 滑铁卢

导师: 沈学民教授 Xuemin (Sherman) Shen

论文: 基于切片的星地融合车联网络资源管理 (Slicing-Based Resource Management for Satellite-Terrestrial Integrated Vehicular Networks)

工程硕士, 电子与通信工程

2017 年 9 月 – 2020 年 3 月

上海交通大学, 中国, 上海

导师: 化存卿教授

论文: 面向双连接网络的并发传输技术研究

交换项目

2017 年 9 月 – 2018 年 2 月

国立巴黎高等矿业学校，法国，巴黎

工学学士，信息工程

2013 年 9 月 – 2017 年 8 月

上海交通大学，中国，上海

毕业设计：异构融合无线网络技术研究

教学经历

课程助教

2025 年 9 月 – 2025 年 12 月

滑铁卢大学

研究生课程：随机过程（ECE604）

助教证书培训研讨会

2023 年 9 月

Certificate of Expectations Teaching Assistant (TA) Training Workshop

滑铁卢大学

研讨会包括：

- 四场不同话题组成的简短讲座
- 在三位学院导师的指导下进行的评分练习

课程助教

2019 年 2 月 – 2019 年 6 月

上海交通大学

本科生课程：数据结构（CS137）

课程助教

2018 年 9 月 – 2019 年 1 月

上海交通大学

本科生课程：C 程序与算法设计（CS139）

研讨会和演讲

- [P1] Resource Slicing with Cross-Cell Coordination in Satellite-Terrestrial Integrated Networks
IEEE ICC 2024 — IEEE International Conference on Communications, 美国丹佛，2024 年 6 月
- [P2] Learning-Based Cache Placement and Content Delivery for Satellite-Terrestrial Integrated Networks
IEEE GLOBECOM 2021 — IEEE Global Communications Conference, 线上会议，2021 年
- [P3] Design and Analysis for Dual Connectivity and Raptor Codes Assisted Handover in Vehicular Networks
IEEE WCNC 2020 — IEEE Wireless Communications and Networking Conference, 线上会议，2020 年 5 月
- [P4] Global Traffic State Recovery VIA Local Observations with Generative Adversarial Networks
IEEE ICASSP 2020 — IEEE International Conference on Acoustics, Speech and Signal Processing, 线上会议，2020 年 5 月

荣誉与奖项

- 最佳论文奖 (第一作者), IEEE WCSP 2025
- 最佳论文奖 (第一作者), IEEE GLOBECOM 2024
- 四分钟论文 (4MT) 竞赛第二名, IEEE 通信协会 2024
- 工程学院奖, 滑铁卢大学, 2021、2024
- 滑铁卢大学研究生奖学金, 滑铁卢大学, 2023 – 2024
- Xuemin Shen 通信领域研究生奖学金, 滑铁卢大学, 2022、2023
- Jon W. Mark, 通信领域研究生奖学金, 滑铁卢大学, 2021
- 加拿大安大略省研究与发展挑战基金 (ORDCF) 贝尔奖学金, 滑铁卢大学, 2021 – 2024
- 国际博士生奖, 滑铁卢大学, 2020 – 2023
- 华为研究生入学奖, 滑铁卢大学, 2020
- 研究生学业奖学金 (一等), 上海交通大学, 2019
- 上海交通大学优秀毕业生, 2017
- 上海交通大学巴黎卓越工程师学院优秀本科毕业设计, 2017
- 上海交通大学学业优秀奖学金 (二等), 2016

论文发表

总引用次数: 436, h 指数: 9

学术著作

- [B1] **M. He**, H. Wu, and X. Shen, “Resource Management for Satellite-Terrestrial Vehicular Networks,” **SpringerBriefs in Computer Science**, Springer, 2026.
Softcover ISBN: 978-3-032-24525-0, eBook ISBN: 978-3-032-24526-7

期刊论文

- [J1] **M. He**, H. Wu, C. Zhou, S. Hu, X. Shen, and W. Zhuang, “Adaptive Multi-Dimensional Resource Slicing in Cognitive Satellite-Terrestrial Vehicular Networks,” **IEEE Transactions on Cognitive Communications and Networking**, vol. 12, pp. 3191-3208, 2026.
- [J2] **M. He**, H. Wu, X. Shen, and W. Zhuang, “Cooperative Resource Scheduling for Environment Sensing in Satellite-Terrestrial Vehicular Networks,” **IEEE Internet of Things Journal**, vol. 12, no. 6, pp. 6734-6748, 2025.
- [J3] **M. He**, C. Hua, W. Xu, P. Gu, and X. Shen, “Delay Optimal Concurrent Transmissions with Raptor Codes in Dual Connectivity Networks,” **IEEE Transactions on Network Science and Engineering**, vol. 8, no. 2, pp. 1478-1491, 2021.

- [J4] J. Li, **M. He***, C. Zhou, X. Huang, Z. Liu, L. Zhao, C.-X. Wang, and H. Wu, “Integration of Generative AI and Mobile Networking: A Comprehensive Survey,” (invited paper), **IEEE Transactions on Network Science and Engineering**, vol. 13, pp. 4369-4405, 2026. (通讯作者)
- [J5] Y. Pei, **M. He**, H. Wu, X. Huang, C. Zhou, W. Zhuang, and X. Shen, “Mobility-Aware Resource Provisioning for Edge-Assisted Extended Reality Services,” **IEEE Internet of Things Journal**, vol. 13, no. 2, pp. 2331-2344, 2026.
- [J6] S. Hu, J. Gao, X. Qin, C. Zhou, X. Huang, M. Li, **M. He**, and X. Shen, “Drift-Adaptive Slicing-Based Resource Management for Cooperative ISAC Networks,” **IEEE Transactions on Cognitive Communications and Networking**, vol. 12, pp. 2631-2646, 2026.
- [J7] M. Ouyang, J. Liu, K. Qu, **M. He**, C. Zhou, B. Wang, R. Zhang, and W. Zhuang, “Two-Level ISTN Intelligent Resource Management: Macro Multipath Planning and Micro A3C-Based Scheduling,” **IEEE Transactions on Network Science and Engineering**, vol. 13, pp. 8517-8534, 2026.
- [J8] J. Yu, C. Hua, L. Liu, P. Gu, and **M. He**, “Accurate Beam Tracking for Robust USV-to-Satellite Transmission under Wave Fluctuation,” **IEEE Transactions on Wireless Communications**, vol. 25, pp. 16320-16335, 2026.
- [J9] Z. Gong, J. Yu, P. Gu, L. Liu, **M. He**, and C. Hua, “Virtual Network Embedding and Dynamic Traffic Scheduling for LEO Satellite Networks,” **IEEE Transactions on Network and Service Management**, submitted, 2025.
- [J10] X. Huang, H. Yang, C. Zhou, **M. He**, X. Shen, and W. Zhuang, “When Digital Twin Meets Generative AI: Intelligent Closed-Loop Network Management,” **IEEE Network**, vol. 39, no. 5, pp. 272-279, 2025.
- [J11] H. Wu, **M. He**, X. Shen, W. Zhuang, N.-D. Dao, and W. Shi, “Network Performance Analysis of Satellite-Terrestrial Vehicular Network,” **IEEE Internet of Things Journal**, vol. 11, no. 9, pp. 16829-16844, 2024.
- [J12] X. Shen, J. Gao, C. Zhou, S. Hu, **M. He**, and W. Zhuang, “Toward Immersive Communications in 6G,” (invited paper), **Frontiers in Computer Science**, vol. 4, pp. 1-31, 2023.
- [J13] Z. Wang, H. Zhu, **M. He**, Y. Zhou, X. Luo, and N. Zhang, “GAN and Multi-Agent DRL Based Decentralized Traffic Light Signal Control,” **IEEE Transactions on Vehicular Technology**, vol. 71, no. 2, pp. 1333-1348, 2021.

会议论文

- [C1] **M. He**, Y. Pei, S. Hu, Z. Tang, W. Zhuang, and X. Shen, “Adaptive Resource Provisioning in Satellite Networks via VAE-Assisted Contextual Bandit Learning,” **WCSP 2025** — International Conference on Wireless Communications and Signal Processing, Chongqing, China, 2025. (最佳论文奖)
- [C2] **M. He**, H. Wu, C. Zhou, S. Hu, Z. Tang, and W. Zhuang, “Digital Twin-Assisted Robust and Adaptive Resource Slicing in LEO Satellite Networks,” **GLOBECOM 2024** — IEEE Global Communications Conference, Cape Town, South Africa, 2024. (最佳论文奖)

- [C3] **M. He**, H. Wu, C. Zhou, and X. Shen, “Resource Slicing with Cross-Cell Coordination in Satellite-Terrestrial Integrated Networks,”
ICC 2024 — IEEE International Conference on Communications, Denver, CO, USA, 2024.
- [C4] **M. He**, C. Zhou, H. Wu, and X. Shen, “Learning-Based Cache Placement and Content Delivery for Satellite-Terrestrial Integrated Networks,”
GLOBECOM 2021 — IEEE Global Communications Conference, Madrid, Spain, 2021.
- [C5] **M. He**, C. Hua, and P. Gu, “Design and Analysis for Dual Connectivity and Raptor Codes Assisted Handover in Vehicular Networks,”
WCNC 2020 — IEEE Wireless Communications and Networking Conference, Seoul, South Korea, 2020.
- [C6] **M. He**, X. Luo, Z. Wang, F. Yang, H. Qian, and C. Hua, “Global Traffic State Recovery via Local Observations with Generative Adversarial Networks,”
ICASSP 2020 — IEEE International Conference on Acoustics, Speech and Signal Processing, Barcelona, Spain, 2020.
- [C7] X. Huang, X. Qin, Y. Zhou, Y. Zhang, **M. He**, J. Li, W. Zhuang, and X. Shen, “Physics-Conditional Diffusion for High-Fidelity Channel Digital Twins in Non-Terrestrial Networks,”
ICCC 2026 — IEEE/CIC International Conference on Communications in China, Wuhan, China, 2026.
- [C8] Y. Pei, **M. He**, S. Hu, H. Hashida, W. Zhuang, and X. Shen, “Joint Rendering Quality and Encoding Type Selection for Edge-Assisted Extended Reality,”
ICC 2026 — IEEE International Conference on Communications, Glasgow, UK, 2026.
- [C9] Z. Gong, J. Yu, S. Hu, P. Gu, L. Liu, **M. He**, and C. Hua, “Virtual Network Embedding based Traffic Scheduling for LEO Satellite Constellations,”
GLOBECOM 2025 — IEEE Global Communications Conference, Taipei, Taiwan, 2025.
- [C10] X. Shao, Y. Zhang, S. Hu, Z. Tang, **M. He**, X. Huang, W. Zhuang, and X. Shen, “Hybrid-Field 6D Movable Antenna for Terahertz Communications: Channel Modeling and Estimation,”
ICCC 2025 — IEEE/CIC International Conference on Communications in China, Shanghai, China, 2025.
- [C11] Y. Pei, S. Hu, C. Zhou, X. Huang, **M. He**, W. Zhuang, and X. Shen, “Interaction-Aware Resource Reservation for Edge-Assisted Mobile Extended Reality,”
ICCC 2025 — IEEE/CIC International Conference on Communications in China, Shanghai, China, 2025.
- [C12] S. Hu, J. Gao, X. Huang, C. Zhou, **M. He**, and X. Shen, “Model Drift-Adaptive Resource Reservation in ISAC Networks: A Digital Twin-Based Approach”
ICCC 2024 — IEEE/CIC International Conference on Communications in China, Hangzhou, China, 2024.
- [C13] C. Zhou, H. Wu, **M. He**, W. Wu, N. Cheng, and X. Shen, “Adaptive Access Mode Selection in Space-Ground Integrated Vehicular Networks,”
GLOBECOM 2021 — IEEE Global Communications Conference, Madrid, Spain, 2021.
- [C14] X. Gong, Y. Zhou, Y. Bi, **M. He**, S. Sheng, H. Qiu, R. He, and J. Lu, “Estimating Web Attack Detection via Model Uncertainty from Inaccurate Annotation,”
CSCloud/EdgeCom 2019 — IEEE International Conference on Cyber Security and Cloud Computing/IEEE International Conference on Edge Computing and Scalable Cloud, Paris, France, 2019.

科研项目

- **Optimization Toolbox Development and Verification for LEO-CAV Networks**
主要参与人, Huawei University Research Program, Huawei Technologies Canada, 2020 – 2023.
- **Intelligent and Customized Service Provisioning in 6G Networks with End-Edge-Cloud Collaboration**
主要参与人, Discovery Grant Program, Natural Sciences and Engineering Research Council of Canada, 2023 – 2027.

志愿经历

- 小组协调人, BCCR 实验室网络小组 (成员：13 人) 2025 年 6 月 – 至今
组织小组成员开展前沿研究，与小组成员讨论科研想法，并组织每周组会。
- 会议志愿者, IEEE PIMRC 2023, 加拿大，多伦多 2023 年 9 月 5 日 – 7 日

专业服务

技术程序委员会

IEEE Global Communications Conference (GLOBECOM) 2026
International Conference on Wireless Communications and Signal Processing (WCSP) 2025
IEEE Global Communications Conference (GLOBECOM) 2025
IEEE International Conference on Computing, Networking and Communications (ICNC) 2025
IEEE Global Communications Conference (GLOBECOM) 2024
IEEE Vehicular Technology Conference Fall (VTC-Fall) 2024
IEEE Vehicular Technology Conference Fall (VTC-Fall) 2023
IEEE International Conference on Computer Communications Workshop (INFOCOM Workshop) 2022

期刊审稿人

IEEE Journal on Selected Areas in Communications (JSAC)
IEEE Transactions on Wireless Communications (TWC)
IEEE Transactions on Mobile Computing (TMC)
Nature Communications
IEEE Transactions on Cognitive Communications and Networking (TCCN)
IEEE Transactions on Vehicular Technology (TVT)
IEEE Transactions on Aerospace and Electronic Systems (TAES)
IEEE Internet of Things Journal (IoTJ)
IEEE Open Journal of the Communications Society (OJCOMS)
IEEE Communications Magazine
IEEE Access
IEEE Network
Springer Peer-to-Peer Networking and Applications
Wireless Networks
China Communications
Electronics and Telecommunications Research Institute (ETRI) Journal

会议审稿人

IEEE Global Communications Conference (GLOBECOM)

IEEE International Conference on Communications (ICC)

IEEE Vehicular Technology Conference (VTC)

IEEE International Conference on Distributed Computing Systems (ICDCS)

IEEE International Conference on Computer Communications Workshop (INFOCOM Workshop)

IEEE Vehicular Technology Conference Workshop (VTC Workshop)

IEEE/CIC International Conference on Communications in China (ICCC)

IEEE Wireless Communications and Networking Conference (WCNC)

IEEE Annual International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC)

IEEE Canadian Conference on Electrical and Computer Engineering (CCECE)

IEEE International Performance Computing and Communications Conference (IPCCC)

IEEE International Conference on Computing, Networking and Communications (ICNC)

IEEE International Conference on High Performance Computing and Communications (HPCC)

IEEE International Conference on Communication Technology (ICCT)

IEEE/CIC International Conference on Communications in China Workshop (ICCC Workshop)